# Quantitative Marketing Research

## From Business Questions to Data-Driven Answers

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# Imagine You Are the Marketing Manager...

1. **Nike's social media team** notices that their posts with the highest number of impressions seem to get lower engagement rates. Should they stop trying to maximize reach?

2. **Apple's brand manager** wants to know: do posts about Product topics drive more engagement than posts about Pricing or Support?

3. **Coca-Cola vs. Pepsi:** A VP asks – "Is there a real difference in how people engage with our brand versus the competitor, or is it just random noise?"

# Why Do We Need Numbers? (Part 1)

- Intuition is valuable but often **wrong** at scale
- Quantitative research gives us three superpowers:
  - **Measure** effect sizes precisely
  - **Compare** alternatives with confidence
  - **Separate** real patterns from noise
- Marketing decisions involve millions of euros

# Why Do We Need Numbers? (Part 2)

## Example: When Intuition Fails

**Your Intuition:**

*"I think negative posts get more likes."*

**What the Data Shows:**

*Positive posts actually get the highest engagement rate (**0.307** vs. **0.262** for positive).*

*This is why we need numbers – our intuitions are often wrong.*

# The Research Process

Business Problem → Research Question → Data Collection → Analysis → Decision

| Step | Example |
|---|---|
| Business Problem | "Our social media engagement is stagnating" |
| Research Question | "Which factors explain variation in engagement rate?" |
| Data | **12,000 posts** across **10 brands** and **5 platforms** |
| Analysis | Correlation, regression, hypothesis tests |
| Decision | Reallocate budget toward high-engagement topics |

# Variance – How Spread Out Is the Data?

- **Variance** measures how much data points differ from the average
- Low variance = clustered tightly; High variance = spread out
- Formula:

$$\text{Variance} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

- **Why it matters:** If there is no variance, there is nothing to explain

# Correlation – Do Two Variables Move Together?

- Correlation measures the **direction** and **strength** of a linear relationship between two numerical variables
- Ranges from **-1** to **+1**:
  - **+1** = perfect positive (as X goes up, Y goes up)
  - **0** = no linear relationship
  - **-1** = perfect negative (as X goes up, Y goes down)
- Notation: $r$ (sample correlation) or $\rho$ (population correlation)
- Important: correlation measures **LINEAR** relationships only

| -1.0 | -0.5 | 0 | +0.5 | +1.0 |
|---|---|---|---|---|
| Perfect Negative | Moderate Negative | No Correlation | Moderate Positive | Perfect Positive |

# How Is Correlation Calculated?

**Pearson correlation coefficient (r):**

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2} \cdot \sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

In plain English:

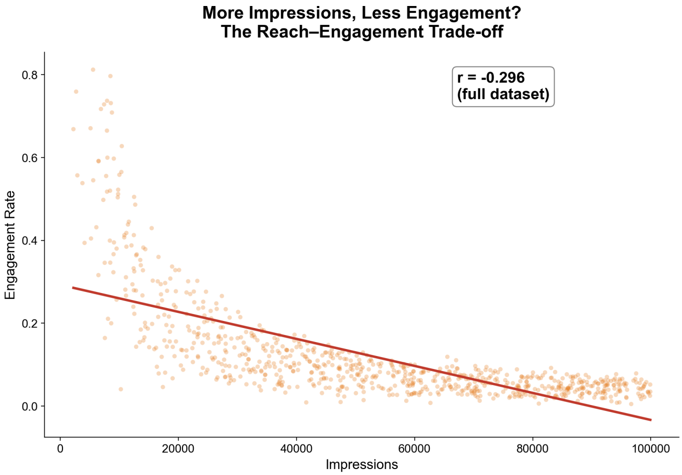
- Measures how much X and Y vary **together** relative to how much they vary **separately**
- Ranges from -1 (perfect negative) to +1 (perfect positive)
- r = 0 means no linear relationship

*You don't need to memorize this formula – software calculates it for you.*
*What matters is understanding what the number means.*

# Positive Correlation Example

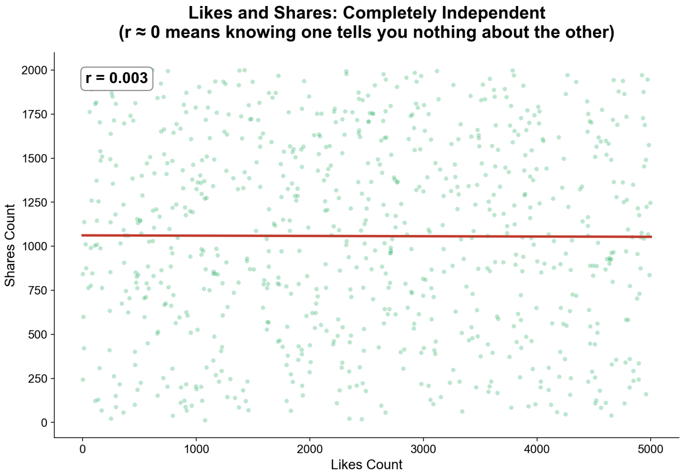

*Posts with higher ad spend tend to get higher sales – a strong positive correlation.*

# Negative Correlation: Impressions and Engagement Rate




*Posts that reach **MORE** people tend to have **LOWER** engagement rates ($r = -0.30$).*

# No Correlation: Likes and Shares




*Knowing how many likes a post got tells you **nothing** about how many shares it will get ($r = 0.003$).*

# Correlation Does Not Mean Causation (Part 1)

- Just because two variables move together does not mean one causes the other
- Three possible explanations for a correlation between X and Y:
  1. **X causes Y** (direct causation)
  2. **Y causes X** (reverse causation)
  3. **Z causes both X and Y** (confounding / spurious correlation)
- Classic example: ice cream sales and drowning rates both go up in summer (confounded by temperature)
- **More hilarious examples:** `https://www.tylervigen.com/spurious-correlations`

# Correlation Does Not Mean Causation (Part 2)

**In our data:** if we find that sentiment_score correlates with engagement_rate, is it because:

1. **Positive sentiment drives engagement?** (X causes Y)
2. **High engagement makes people write more positively?** (Y causes X)
3. **Certain brands/topics produce both positive sentiment AND high engagement?** (confounder Z)

*We need regression with controls to untangle these possibilities.*

# Controlling for Confounders

- A **control variable** is a factor we hold constant to isolate the relationship we care about
- Example: To study the effect of sentiment on engagement, we need to control for brand and platform
  - Without controls: sentiment and engagement might correlate simply because both are driven by brand
  - With controls: we compare posts from the **same brand, same platform** and ask – among those, does sentiment still matter?
- This is the bridge from correlation to regression

*Holding brand constant, does sentiment still predict engagement?*

# Linear Regression – Finding the Best-Fitting Line

**Regression equation:**

$$Y = \alpha + \beta X + \varepsilon$$

- $\alpha$ = intercept
- $\beta$ = slope (key number!)
- $\varepsilon$ = error

"**Best-fitting**" = line closest to all data points

## The Slope (Beta) – The Key Number

- The slope $\beta$ (also called beta) is the most important output of a regression
- Interpretation: "For every 1-unit increase in X, Y changes by $\beta$ units"
- The sign of $\beta$ tells direction:
  - $\beta > 0$: positive relationship
  - $\beta < 0$: negative relationship
  - $\beta = 0$: no relationship

*"For every one-unit increase in X, Y changes by $\beta$ units, on average."*

# Interpreting Beta: An Example

**Example with our data:**

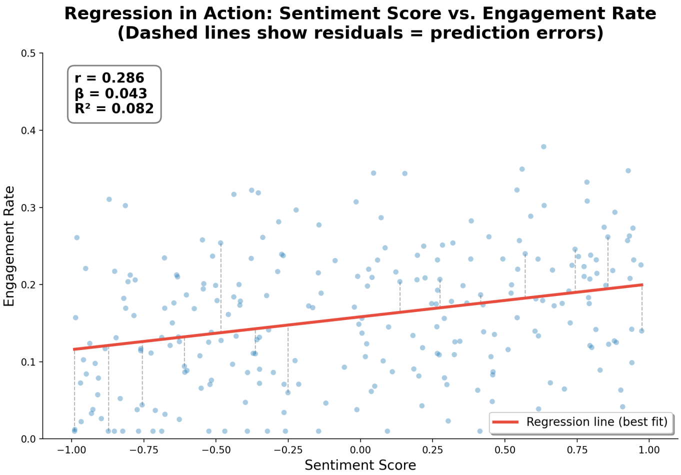
If we regress `engagement_rate` on `sentiment_score`:

$$\beta = 0.042$$

**Interpretation:** A **1-point increase** in sentiment (e.g., from 0.0 to 1.0) is associated with a **0.042 increase** in engagement rate

*This is the "effect size" – what marketers need for resource allocation decisions.*

# Regression in Action: Sentiment Predicting Engagement Rate




*The red line is the regression line. Vertical distances = residuals = prediction errors.*

# R-Squared – How Much Variance Does the Model Explain?

- R-squared ($R^2$) measures the proportion of variance in Y explained by the model
- Ranges from **0 to 1**:
  - $R^2 = 0$: the model explains **nothing** (the line is useless)
  - $R^2 = 1$: the model explains **everything** (all points on the line)
  - $R^2 = 0.08$ (for impressions predicting engagement): the model explains **8% of the variance**
- That means **92%** of the variation in engagement rate is due to **OTHER factors**

# Multiple Regression – Including Control Variables

- We can add multiple predictors to the model:

$$\text{engagement\_rate} = \alpha + \beta_1 \cdot \text{sentiment} + \beta_2 \cdot \text{brand} + \beta_3 \cdot \text{platform} + \varepsilon$$

- Each $\beta$ coefficient now represents the effect of that variable **holding all others constant**
  - $\beta_1$ = effect of sentiment on engagement, **controlling for** brand and platform
- This addresses the confounding problem from earlier
- Categorical variables (brand, platform) are included as "dummy variables" – the software handles this

*Each coefficient = the effect of that variable, holding all others constant*

# What Predicts Engagement Rate? (Multiple Regression Results)

| Predictor | Coefficient ($\beta$) | Interpretation |
|---|---|---|
| sentiment_score | 0.03 | Positive effect, controlling for all else |
| Brand: Nike | 0.15 | Nike's engagement premium vs. baseline |
| Topic: Product | 0.08 | Product topic premium vs. baseline |
| $R^2$ | 0.58 | |

*The full model explains more variance than any single predictor alone.*

# Population vs. Sample

- **Population:** All social media posts about these **10 brands** (**millions**)
- **Sample:** Our dataset of **12,000 posts**
- We use the sample to make **inferences** about the population
- Every estimate has **uncertainty**

*The dataset is a window into the full population.*

# Example: Estimating Average Height in Spain

**Research Question:** What is the average height of university students in Spain?

**What we do:**

- We measure **10 students** in this classroom
- We calculate the **average height** of these 10 students
- We use this to **estimate** the population average

**Heights (cm): 172, 168, 175, 165, 180, 171, 169, 177, 173, 170**

*Can we trust this sample of 10 to tell us about millions of students?*

# Calculating the Confidence Interval

**Step 1: Calculate the sample mean**

$$\bar{x} = \frac{172 + 168 + 175 + 165 + 180 + 171 + 169 + 177 + 173 + 170}{10} = 172 \textbf{ cm}$$

**Step 2: Calculate the standard deviation**

$$s = 4.5 \textbf{ cm}$$

**Step 3: Calculate the standard error**

$$SE = \frac{s}{\sqrt{n}} = \frac{4.5}{\sqrt{10}} = 1.4 \textbf{ cm}$$

**Step 4: Build the 95% confidence interval**

$$\text{CI}_{95\%} = \bar{x} \pm 1.96 \times SE = 172 \pm 1.96 \times 1.4 = 172 \pm 2.7$$

$$\text{CI}_{95\%} = [169.3, 174.7] \textbf{ cm}$$

# What Does This Confidence Interval Mean?

**Our 95% confidence interval: [169.3, 174.7] cm**

**Interpretation:**

- We are **95% confident** that the **true average height** of Spanish university students is between **169.3 and 174.7 cm**
- If we repeated this experiment **100 times** (measuring 10 different students each time), about **95 of the intervals** would contain the true population mean
- The interval is **wide** (**±2.7 cm**) because our sample is **small** ($n = 10$)

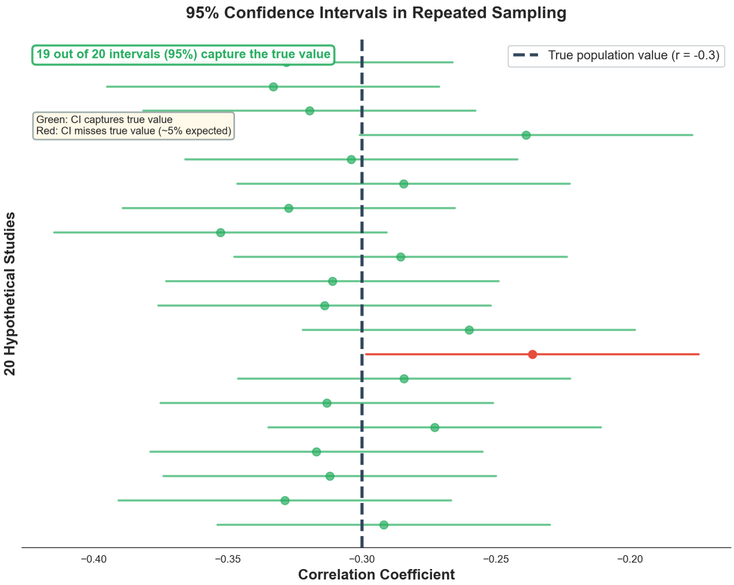
*Larger samples → narrower intervals → more precision*

# How Certain Are We About Our Estimates?

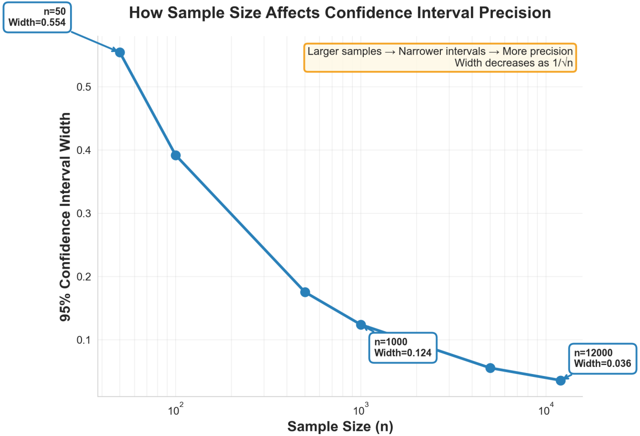
- Every statistic is an **estimate** with **uncertainty**
- **Confidence interval:** "We are 95% confident the true value is between X and Y"
- Narrow interval = high precision; Wide interval = low precision
- Formula: $\text{CI} = \text{estimate} \pm 1.96 \times \frac{s}{\sqrt{n}}$
  - $s$ = standard deviation, $n$ = sample size

*If we repeated the study **100 times**,*
***95 intervals** would contain the true value*

# Confidence Intervals: Visual Interpretation



# Why Sample Size Matters: Precision of Estimates




- Larger sample $\rightarrow$ narrower CI $\rightarrow$ more precision
- CI width $\propto 1/\sqrt{n}$

# Example: How Much Does Sentiment Affect Engagement?

- From our regression: engagement_rate $= \alpha + \beta \times$ sentiment_score + ...
- Point estimate: $\beta = 0.042$
  - Interpretation: A **1-unit** increase in sentiment is associated with a **0.042 increase** in engagement rate
- **95% Confidence Interval: [0.028, 0.056]**
  - Interpretation: We are **95% confident** the true effect is between **0.028 and 0.056**
- Since the interval does **NOT include zero**, we can be confident sentiment has a **real, positive effect**
- This is the visual counterpart to hypothesis testing (which we will cover next)

## The Fundamental Problem: Is It Real or Is It Noise?

- We found $\beta = 0.042$ for the effect of sentiment on engagement
- Does this exist in the **population**, or just our **sample**?
- Hypothesis testing answers: **Is this real or random noise?**

*Quantifying confidence with a known error rate*

# Setting Up Hypotheses

**Null Hypothesis ($H_0$):** There is **no effect** / no relationship / no difference

- "Sentiment score has no effect on engagement ($\beta = 0$)"
- "There is no difference in engagement between Nike and Adidas"
- "Brand has no effect on engagement"

**Alternative Hypothesis ($H_1$ or $H_a$):** There **is** an effect / relationship / difference

- "Sentiment DOES affect engagement ($\beta \neq 0$)"
- "There IS a difference between Nike and Adidas"
- "Brand DOES affect engagement"

*We start by assuming $H_0$ is true and ask: how likely is our data under that assumption?*

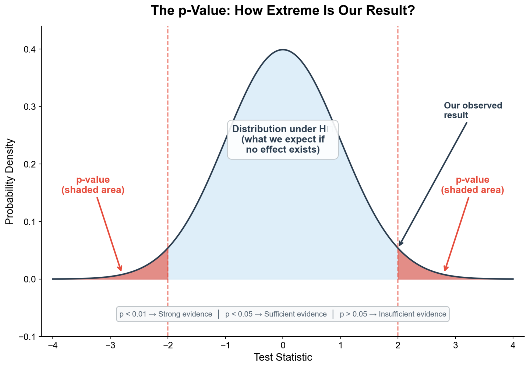
# How Hypothesis Testing Works

**The logic, step by step:**

1. **Assume $H_0$ is true** (no effect exists)
2. **Ask:** If $H_0$ were true, how likely is it that we would observe data as extreme as what we got?
3. **Calculate this probability** – this is the **p-value**
4. **Decision rule:**
   - If p-value is very small (typically $< 0.05$): the data is very unlikely under $H_0 \rightarrow$ **reject** $H_0$
   - If p-value is not small ($\geq 0.05$): we cannot rule out $H_0 \rightarrow$ **fail to reject** $H_0$

*If the p-value is below **0.05**, we reject the null and conclude there **IS** an effect.*

# What Is a p-Value?




**p-value** $=$ probability of seeing data this extreme *if $H_0$ were true*

$p < 0.01$: **Strong evidence** against $H_0$

$p < 0.05$: **Sufficient evidence** (conventional threshold)

$p > 0.05$: **Insufficient evidence**

# Example: Does Sentiment Really Affect Engagement?

- $H_0$: $\beta = 0$ (sentiment has **no effect** on engagement in the population)
- $H_1$: $\beta \neq 0$
- In our regression: $\beta = 0.042$, $n = 12{,}000$
- Standard error: $SE = 0.008$
- Test statistic: $t = \frac{\beta}{SE} = \frac{0.042}{0.008} = 5.25$
- p-value: **extremely small** ($p < 0.001$)

**Conclusion: REJECT $H_0$**
The positive effect of sentiment on engagement is statistically significant.
This is very unlikely to be random noise.

# Two Types of Mistakes

| | $H_0$ is TRUE (no effect) | $H_0$ is FALSE (effect exists) |
|---|---|---|
| We reject $H_0$ | TYPE I ERROR (False Positive) | CORRECT (True Positive) |
| We fail to reject $H_0$ | CORRECT (True Negative) | TYPE II ERROR (False Negative) |

- **Type I Error ($\alpha$):** False positive (accept 5% risk)
- **Type II Error ($\beta$):** False negative (depends on sample size)
- **Power** $= 1 - \beta =$ probability of detecting a real effect

# Real Consequences: Type I Error (False Positive)

**Scenario:** We test whether the "Product" topic gets more engagement than "Support" topics

**Type I Error:** We conclude Product topics get more engagement, but it is actually just noise

**Consequence:**

- The brand reallocates its content budget toward Product topics
- Money is wasted because there was no real difference
- The Support team loses resources for no reason

*Acting on a false positive wastes budget and resources.*

# Real Consequences: Type II Error (False Negative)

**Scenario:** We test whether sentiment score predicts engagement

**Type II Error:** We conclude sentiment has no effect, but it actually does

**Consequence:**

- The brand ignores sentiment management
- Negative sentiment goes unchecked, quietly eroding engagement over months
- An opportunity to improve engagement by managing tone is missed

*Missing a real effect means missed opportunities.*

# Statistical vs. Practical Significance (Part 1)

- With $n = 12{,}000$, even **tiny effects** become statistically significant
- Example: sentiment_score vs. toxicity_score: $r = 0.008$
  - With **12,000 observations**, even $r = 0.008$ might be statistically significant ($p < 0.05$)
  - But $r = 0.008$ explains only **0.006% of the variance** – practically meaningless
- This is the **"large sample trap"**: with enough data, everything becomes significant

# Statistical vs. Practical Significance (Part 2)

**Always ask TWO questions:**

1. Is the effect **statistically significant**? ($p < 0.05$?)
   - Is it real, or just noise?
2. Is the effect **practically meaningful**?
   - Is it large enough to matter for business decisions?

*Always pair statistical significance with practical significance.*

# Hypothesis Testing Meets Regression

- Every coefficient in a regression output comes with:
  - The estimate ($\beta$)
  - A standard error (uncertainty in $\beta$)
  - A t-statistic ($\beta$ / standard error)
  - A p-value (is $\beta$ significantly different from zero?)

| **Predictor** | $\beta$ | **Std. Error** | **t-stat** | **p-value** | **Significant?** |
|---|---|---|---|---|---|
| impressions | -0.00003 | 0.000002 | -15.0 | $< 0.001$ | Yes |
| sentiment_score | 0.042 | 0.008 | 5.25 | $< 0.001$ | Yes |
| Brand: Nike | 0.15 | 0.035 | 4.29 | $< 0.001$ | Yes |

*For each predictor, $H_0$ says $\beta = 0$ (no effect). If $p < 0.05$, we reject $H_0$ and conclude the effect is real.*

# Back to Our Three Questions

1. **Nike's impressions vs. engagement dilemma:**
    - $r = -0.30$, $p < 0.001$ $\rightarrow$ The trade-off is **real and statistically significant**
    - Implication: maximizing reach does **not** maximize engagement. Target better, not broader.
2. **Apple's topic strategy:**
    - Topic category explains **31% of engagement variance**
    - Some topics drive significantly more engagement than others (regression confirms this)
    - Implication: reallocate content toward high-engagement topics
3. **Coca-Cola vs. Pepsi:**
    - Brand explains **50% of engagement variance**
    - Hypothesis test can confirm whether the specific Coca-Cola vs. Pepsi difference is significant
    - Implication: brand identity is the **single biggest driver** of engagement

# What You Should Remember

**Five key takeaways:**

1. **Variance** is the raw material – if nothing varies, there is nothing to study
2. **Correlation** tells you whether two variables move together (and how strongly), but NOT whether one causes the other
3. **Regression** draws the best-fitting line, quantifies effects, and lets you control for confounders
4. **R-squared** tells you how much of the outcome your model explains
5. **Hypothesis testing** tells you whether your findings are real or just noise – but always pair statistical significance with practical significance

*The toolkit: Variance → Correlation → Regression → Hypothesis Testing*

# Questions?

## Thank You!

**Dataset:** Social Media Engagement Dataset
(12,000 posts, 10 brands, 5 platforms)

**Tools covered:** Correlation, Linear Regression, R-squared,
Confidence Intervals, Hypothesis Testing

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